

A
Real-Time Research Project
On
**IMAGE STEGANOGRAPHY WITH CNN BASED ENCODER-DECODER
MODEL**

(Submitted in partial fulfilment of the requirements for the award of Degree)

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CERTIFICATE

This is to certify that the project entitled “**IMAGE STEGANOGRAPHY WITH CNN BASED ENCODER-DECODER MODEL**” being submitted by Rishith Kumar Shetty (247R1A05B8), Gurralla Sai Sudha (247R1A0586) , Lokam Shruthilaya (247R1A0599) in partial fulfillment of the requirements for the award of the degree of B.Tech in Computer Science and Engineering to the Jawaharlal Nehru Technological University Hyderabad, during the year 2025-26.

The results embodied in this thesis have not been submitted to any other University or Institute for the award of any degree or diploma.

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ABSTRACT

Image steganography is a significant area of research that aims at finding any hidden images or information within other digital images. In this paper, we propose a novel approach combining the Convolutional Neural Network (CNN) model and an encoder and decoder network to ensure accurate and efficient image steganalysis. The proposed method leverages the discriminative power of CNNs to extract features of importance from images and then uses the encoder-decoder networks to reconstruct the original image from the stego or hidden content. Our customized CNN model is designed to capture the features present in the steganographic images. The Encoder-Decoder network plays a crucial role in steganalysis by reconstructing the original image from the hidden content. By training the network on a diverse set of steganographic images, it learns to identify the distinctive artifacts introduced during the steganographic embedding process. The reconstructed image is then compared with the original image using appropriate similarity measures, allowing us to accurately detect the presence of hidden information. The combination of the customized CNN model and the Encoder-Decoder network enables efficient and robust image steganalysis, making it a valuable tool for digital forensics and security applications.

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1. INTRODUCTION

Image Steganography with CNN Based Encoder-Decoder Mode

1.INTRODUCTION

Image Steganography is a method of hiding information within digital images to ensure secure communication. Leveraging advancements in deep learning, Convolutional Neural Networks (CNNs) are now used to create encoder-decoder models for embedding and retrieving secret messages. This approach enhances the robustness and efficiency of steganography compared to traditional methods. The CNN-based models not only allow precise encoding but also improve the quality of the decoded images.

1.1 PROJECT PURPOSE

The main purpose of this project is to develop a secure and intelligent image steganography system using a Convolutional Neural Network (CNN)–based encoder–decoder architecture. The project aims to enhance secure communication by embedding confidential information within digital images in such a way that the hidden data is invisible to the human eye and difficult to detect by steganalysis techniques.

This project focuses on utilizing deep learning methods to overcome the limitations of traditional steganographic techniques such as Least Significant Bit (LSB) substitution. By leveraging CNNs, the system automatically learns complex image features and identifies optimal embedding locations to minimize visual distortion while maximizing payload capacity. The encoder network embeds the secret message into the cover image, and the decoder network accurately retrieves the hidden information from the stego image

Image Steganography with CNN Based Encoder-Decoder Model

1.2 PROJECT FEATURES

The Image Steganography using CNN Encoder–Decoder project is designed with advanced deep learning capabilities to ensure secure and efficient data hiding within digital images.

The system uses a Convolutional Neural Network (CNN) to automatically extract important image features and identify suitable regions for embedding secret information. The encoder network embeds the confidential message into the cover image while maintaining high visual quality, ensuring that the stego image appears almost identical to the original image. The decoder network accurately retrieves the hidden message by learning subtle variations introduced during the embedding process.

This project provides strong robustness against common image processing attacks such as compression, noise addition, cropping, and filtering. It supports improved payload capacity compared to traditional steganography methods and maintains image similarity using evaluation metrics like Mean Squared Error (MSE) and Structural Similarity Index (SSIM).

The end-to-end training mechanism allows both encoding and decoding processes to be optimized simultaneously, improving overall performance and reliability. Due to its adaptability and high accuracy, the system can be effectively applied in secure communication, digital forensics, cybersecurity, and confidential multimedia data transmission.

2. LITERATURE SURVEY

Image Steganography with CNN Based Encoder-Decoder Model

2. LITERATURE SURVEY

Image steganography has evolved significantly from traditional pixel-based techniques to advanced deep learning-based approaches. Early steganographic methods such as Least Significant Bit (LSB) substitution focused on embedding secret data directly into pixel values.

While these methods were simple and computationally efficient, they were highly vulnerable to statistical and visual attacks. Researchers later introduced transform-domain techniques such as Discrete Cosine Transform (DCT) and Discrete Wavelet Transform (DWT), which improved robustness by embedding information in frequency components rather than raw pixels. However, these approaches still relied heavily on manual feature selection and predefined embedding rules.

With the advancement of deep learning, particularly Convolutional Neural Networks (CNNs), researchers began exploring automated feature extraction for both steganography and steganalysis. CNNs demonstrated strong capability in identifying hidden patterns and subtle distortions in images. One of the early breakthroughs was the application of deep CNN models for steganalysis, which significantly improved detection accuracy compared to traditional handcrafted feature-based methods.

A major milestone in deep learning-based steganography was the work by ShumeetBaluja in the paper *Hiding Images in Plain Sight: Deep Steganography*, where an end-to-end neural network was proposed to hide one image inside another. This model used an encoder-decoder architecture to embed and reconstruct hidden images with minimal perceptual distortion [2].

Further advancements were introduced through Generative Adversarial Networks (GANs), where a generator network embeds secret information and a discriminator attempts to detect it. This adversarial training process improved resistance against steganalysis attacks [3].

Image Steganography with CNN Based Encoder-Decoder Model

Recent studies focus on improving robustness against real-world distortions such as JPEG compression, noise, scaling, and cropping. Hybrid models combining CNNs with residual networks (ResNet) and attention mechanisms have been proposed to enhance feature learning and improve embedding capacity. Additionally, lightweight architectures are being developed to reduce computational complexity and enable real-time applications [5].

Overall, the literature indicates a clear transition from traditional statistical steganography methods to intelligent, data-driven deep learning frameworks. CNN-based encoder-decoder architectures have emerged as a powerful solution due to their ability to automatically learn complex image representations, maintain high visual quality, and achieve strong resistance against detection. These advancements form the foundation for modern secure image communication systems and motivate the development of more robust and efficient steganographic models.

Image steganography has undergone significant transformation over the past two decades. Initially, research focused on spatial domain techniques such as Least Significant Bit (LSB) substitution, pixel value differencing, and edge-based embedding methods. These approaches were simple to implement and offered good payload capacity, but they lacked robustness against statistical attacks and image processing operations.

As a result, researchers shifted towards transform-domain techniques such as Discrete Cosine Transform (DCT) and Discrete Wavelet Transform (DWT), where secret data was embedded into frequency coefficients. These methods improved resistance to compression and filtering but still relied on handcrafted rules and manual feature engineering [1].

The emergence of deep learning marked a turning point in steganography research. Convolutional Neural Networks (CNNs) demonstrated exceptional performance in image classification, feature extraction, and pattern recognition tasks. Researchers recognized that CNNs could automatically learn complex hierarchical features from images, making them highly suitable for both steganography (data hiding) and steganalysis (hidden data detection)[4].

Image Steganography with CNN Based Encoder-Decoder Model

Further advancements were made using Generative Adversarial Networks (GANs). In GAN-based steganography, a generator network embeds secret data while a discriminator network attempts to detect the presence of hidden information.

This adversarial training process strengthens the embedding network by forcing it to produce highly realistic stego images. GAN-based methods significantly improve resistance against steganalysis and enhance security.

Recent literature also emphasizes robustness against real-world distortions such as JPEG compression, resizing, cropping, and noise injection. Residual networks (ResNet), attention mechanisms, and dense connections have been incorporated into CNN-based steganography systems to improve feature propagation and embedding efficiency. Researchers are also working on lightweight architectures to reduce computational complexity and enable deployment in mobile and IoT devices.

Additionally, studies compare performance metrics such as Peak Signal-to-Noise Ratio (PSNR), Mean Squared Error (MSE), and Structural Similarity Index (SSIM) to evaluate image quality and embedding performance. High PSNR and SSIM values indicate better imperceptibility, while low MSE ensures minimal distortion between cover and stegoimages.

In summary, the literature clearly shows a shift from traditional rule-based steganographic techniques to intelligent, data-driven deep learning models. CNN-based encoder-decoder architectures provide automated feature learning, higher payload capacity, improved visual quality, and strong resistance against detection. These advancements form the foundation for modern secure image communication systems and motivate continued research in adaptive, robust, and real-time steganography frameworks for cybersecurity and digital forensics applications.

Image Steganography with CNN Based Encoder-Decoder Model

2.1 REVIEW OF RELATED WORK

A Review of Related Work is a systematic summary and analysis of previously published research, studies, and methodologies related to a particular topic. It examines existing theories, techniques, models, and findings to understand the current state of knowledge in the field. In the context of Image Steganography using CNN Encoder–Decoder, the review of related work analyzes earlier steganography techniques, deep learning approaches, encoder–decoder architectures, and CNN-based models to identify their strengths, limitations, and research gaps.

1. Traditional Spatial Domain Techniques

- Early steganography methods focused on pixel-level embedding.
- Least Significant Bit (LSB) substitution was widely used due to simplicity.
- Provided high payload capacity but low security.
- Vulnerable to statistical and visual steganalysis attacks.
- Easily affected by image compression and noise

2. Transform Domain Techniques

- Data embedded in frequency components using DCT and DWT.
- Improved robustness against JPEG compression and filtering.
- Reduced visible distortion compared to spatial methods.
- Still dependent on handcrafted embedding rules.
- Limited adaptability to new attack techniques.
- Introduction of feature-based classifiers for hidden data detection.
- Manual feature extraction followed by classification algorithms.
- Improved detection accuracy compared to traditional statistical methods.
- Performance depended heavily on selected features.

Machine Learning–Based Steganalysis refers to the application of machine learning algorithms to automatically detect the presence of hidden information within digital images. Unlike traditional statistical methods that rely on fixed rules and predefined thresholds, machine learning approaches use data-driven models trained on large datasets of cover (original) and stego (modified) images

Image Steganography with CNN Based Encoder-Decoder Model

These models learn patterns and statistical irregularities introduced during the embedding process, enabling more accurate and adaptive detection.

4. CNN-Based Steganalysis

- Convolutional Neural Networks (CNNs) introduced automatic feature extraction.
- Capable of detecting subtle embedding artifacts.
- Achieved higher accuracy compared to traditional machine learning models.
- Eliminated need for manual feature engineering.
- Marked the transition to deep learning approaches.

5. Deep Learning-Based Steganography

Deep Learning-Based Steganography refers to the use of advanced neural network architectures to automatically learn how to embed and extract secret information within digital images. Unlike traditional steganographic techniques that rely on fixed mathematical rules or handcrafted embedding strategies, deep learning models learn optimal hiding patterns directly from large datasets. This data-driven approach enables the system to adapt to complex image characteristics and achieve higher security, better image quality, and improved robustness.

One of the most widely used architectures in deep learning-based steganography is the encoder-decoder model. In this framework, the encoder network takes a cover image and a secret message as input and produces a stego image that visually resembles the original image. The decoder network then extracts or reconstructs the hidden message from the stego image. Both networks are trained simultaneously in an end-to-end manner using loss functions that balance embedding accuracy and image similarity.

2.2 DEFINITION OF PROBLEM STATEMENT

A Problem Statement is a clear and concise description of the issue that a project aims to address. It defines the existing problem, explains why it is important, and outlines the need for a solution. The problem statement identifies the limitations of current systems or methods and establishes the objectives that the proposed system intends to achieve.

Image Steganography with CNN Based Encoder-Decoder Model

2.3 EXISTING SYSTEM

Traditional image steganography systems rely on methods like Least Significant Bit (LSB) manipulation, Discrete Cosine Transform (DCT), and other signal processing techniques. These approaches have limitations in terms of payload capacity, robustness against attacks, and visual imperceptibility of the embedded image. Additionally, traditional systems often lack the adaptability required to handle the diverse and dynamic challenges in secure communication.

Limitations of Existing System

1. Limited payload capacity, restricting the amount of data that can be embedded.
2. Susceptibility to attacks such as noise addition, compression, or cropping.
3. Degradation in image quality due to simplistic embedding methods.
4. Lack of scalability and adaptability for complex image datasets.

2.4 PROPOSED SYSTEM

The proposed system employs a CNN-based encoder-decoder model to perform image steganography. The encoder is responsible for embedding the secret message into the image, while the decoder retrieves it without compromising the quality of the image or the message. The system leverages the learning capabilities of CNNs to optimize the encoding and decoding processes, ensuring imperceptibility, robustness, and accuracy in various scenarios. By using deep learning, the proposed system adapts to varying image features and enhances security.

Advantages of the Proposed System:

1. High payload capacity for embedding larger messages.
2. Enhanced robustness against various attacks and distortions.
3. Maintains the visual quality of the carrier image.
4. Increased retrieval accuracy with minimal data loss.
5. Adaptive to diverse datasets and dynamic requirements

Image Steganography with CNN Based Encoder-Decoder Model

2.5 OBJECTIVES

1.Input Design is the process of converting a user-oriented description of the input into a computer-based system. This design is important to avoid errors in the data input process and show the correct direction to the management for getting correct information from the computerized system.

2.It is achieved by creating user-friendly screens for the data entry to handle large volume of data. The goal of designing input is to make data entry easier and to be free from errors. The data entry screen is designed in such a way that all the data manipulates can be performed. It also provides record viewing facilities.

3.When the data is entered it will check for its validity. Data can be entered with the help of screens. Appropriate messages are provided as when needed so that the user will not be in maize of instant. Thus the objective of input design is to create an input layout that is easy to follow

2.6 HARDWARE & SOFTWARE REQUIREMENTS

2.6.1 HARDWARE REQUIREMENTS:

Hardware interfaces specify the logical and physical characteristics of each interface between the software system and the hardware components required for implementing **Image Steganography using CNN Encoder–Decoder**. Since the system involves deep learning model training and image processing, moderate computational resources are required.

- System : Pentium Dual Core.
- Hard Disk : 120 GB.
- Monitor : 15’’ LED
- Input Devices : Keyboard, Mouse
- Ram : 1 GB

Image Steganography with CNN Based Encoder-Decoder Model

2.6.2 SOFTWARE REQUIREMENTS:

Software requirements specify the logical characteristics of the software environment and tools required to develop and run the **CNN-based Image Steganography system**.

- Operating system : Windows 10/11.
- Coding Language : python
- Tool : pycharm
- Database : MYSQL
- framework : django

3. SYSTEM ARCHITECTURE & DESIGN

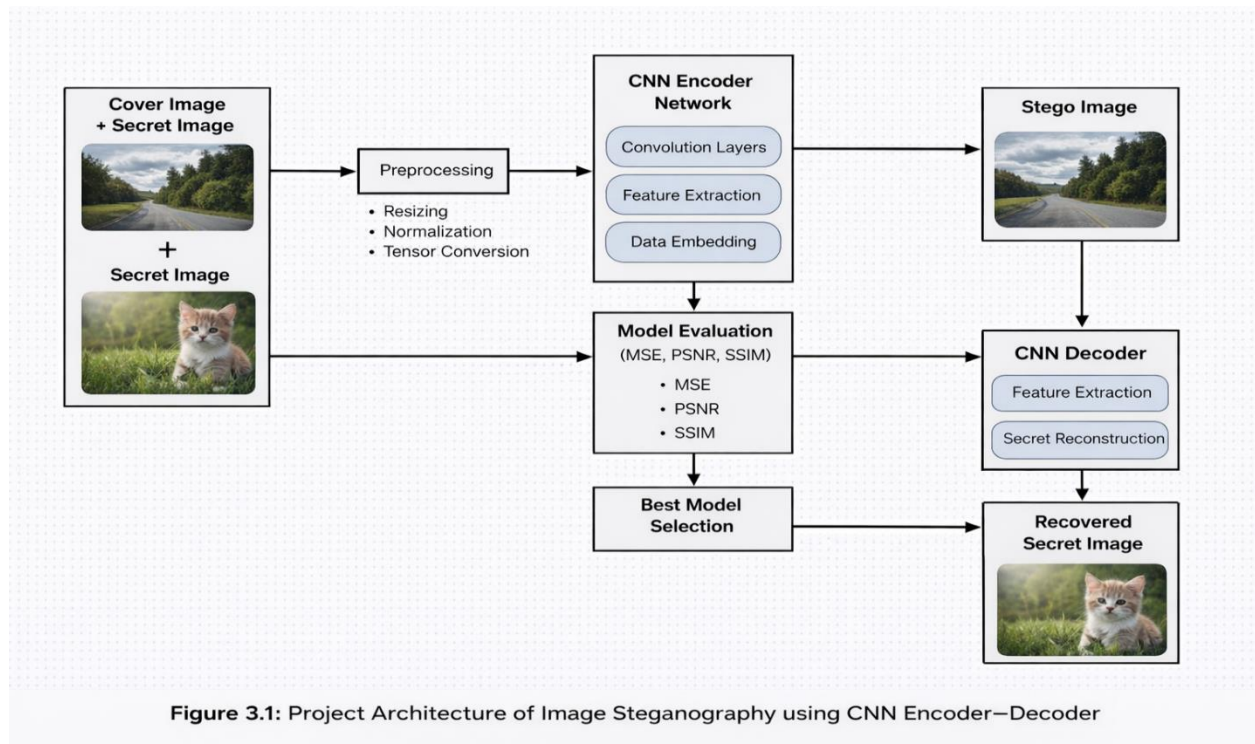
Image Steganography with CNN Based Encoder-Decoder Model

3. SYSTEM ARCHITECTURE & DESIGN

Project architecture defines the structure of the Image Steganography with CNN-Based Encoder-Decoder Model, including components for data preprocessing, secret data embedding, and message extraction while maintaining image quality.

3.1 PROJECT ARCHITECTURE

The project architecture of Image Steganography using CNN Encoder-Decoder describes the overall structure of the system and how different components interact to securely embed and extract secret images within cover images using deep learning techniques.



3.2 DESCRIPTION

Input Data:The images are uploaded by the user through the interface and loaded into the system for further processing.**Feature Extraction :**The Convolutional Neural Network extracts important features from both the cover image and the secret image using convolutional layers. Feature maps are generated to capture patterns, textures, and spatial information

Image Steganography with CNN Based Encoder-Decoder Model

Data Preprocessing :The input images undergo preprocessing to prepare them for the CNN Model

Secret Image Embedding (CNN Encoder) :The encoder network embeds the secret image features into the cover image features.This process generates a **Stego Image** that looks almost identical to the original cover image but contains hidden data.

Stego Image Generation :The output of the encoder is the stegoimage.It maintains high visual quality while securely storing the secret information.

3.3 DATA FLOW DIAGRAM

A Data Flow Diagram (DFD) is a graphical representation that illustrates how data flows through a system. It shows the input, processing stages, data storage (if any), and output of the system. In the proposed system, DFD explains how images move through different stages of the CNN-based steganography model. A Data Flow Diagram comprises Four primary elements:

The stego image is then passed to the CNN Decoder module. The decoder processes the stego image and extracts the hidden features to reconstruct the secret image. This ensures secure transmission and accurate recovery of hidden data.

Benefits:

The proposed system of Image Steganography using a CNN Encoder–Decoder architecture offers several advantages over traditional steganography techniques. By utilizing deep learning, the system automatically learns optimal embedding and extraction strategies, resulting in improved security and performance.

Applications:

- Image Steganography using CNN Encoder–Decoder has wide applications in secure communication and digital data protection. Since the system hides confidential information inside images without noticeable distortion, it can be used in various real-world scenarios.

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- One of the primary applications is **secure communication**. Sensitive information such as personal messages, confidential documents, or private images can be securely transmitted over the internet without revealing the presence of hidden data.

Levels of DFD:

DFDs are structured hierarchically:

- Level 0 (Context Diagram): Level 0 provides a high-level overview of the entire Image Steganography system.
- Level 1: breaks down the main system into major sub-processes involved in steganography
- Level 2+: Level 2 provides deeper insight into specific processes such as the Encoder and Decoder modules

Data Flow Diagram of Image Steganography using CNN Encoder–Decoder

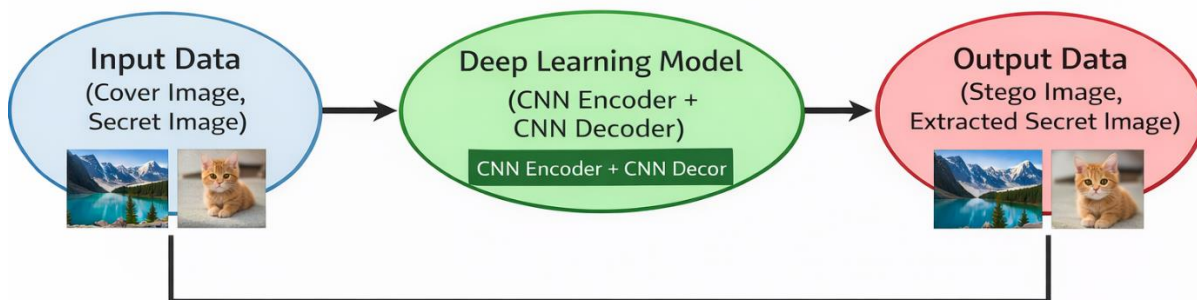


Figure 3.2: Dataflow Diagram of A Image Steganography using CNN Encoder - Decoder

4. IMPLEMENTATION

Image Steganography with CNN Based Encoder-Decoder Model

4.IMPLEMENTATION

The implementation phase of the project “**Image Steganography using CNN Encoder–Decoder**” focuses on developing and integrating deep learning models to securely embed and extract secret information within digital images. The system is designed to ensure high embedding capacity, strong security, and minimal visual distortion in the cover image.

The proposed architecture consists of a **CNN-based Encoder–Decoder framework**, where:

- The **Encoder Network** hides the secret image/message into the cover image.
- The **Decoder Network** extracts the hidden message from the stego image.
- The system is trained end-to-end to minimize visual difference and maximize recovery accuracy.

The implementation includes dataset preparation, preprocessing, model training, optimization, and performance evaluation using metrics such as PSNR (Peak Signal-to-Noise Ratio), SSIM (Structural Similarity Index), and MSE (Mean Square Error).

4.1 ALGORITHMS USED

CNN-Based Encoder–Decoder Model

Convolutional Neural Networks (CNNs) are widely used in image processing tasks due to their strong capability to extract spatial features. In this project, a CNN-based Encoder–Decoder architecture is used for secure image steganography.

The **Encoder Network** embeds the secret image into the cover image by learning complex pixel-level transformations. The **Decoder Network** reconstructs the hidden image from the generated stego image.

Unlike traditional steganography techniques such as LSB (Least Significant Bit), DCT (Discrete Cosine Transform), or DWT (Discrete Wavelet Transform), the CNN-based model automatically learns optimal embedding patterns without manual feature engineering.

Image Steganography with CNN Based Encoder-Decoder Model

Advantages of CNN-Based Encoder–Decoder Model:

- Automatically learns complex embedding patterns.
- Provides higher payload capacity compared to traditional methods.
- Produces stego images with minimal visible distortion.
- More robust against steganalysis attacks.
- End-to-end learning improves reconstruction accuracy.

Disadvantages of CNN-Based Encoder–Decoder Model:

- Requires large datasets for effective training.
- Computationally expensive during training.
- Requires GPU for faster performance.
- Performance may degrade if trained on limited image diversity.

2. Encoder Network for Secret Image Embedding

The Encoder is responsible for hiding the secret image into the cover image.

- Takes two inputs: Cover Image and Secret Image.
- Uses convolutional layers to extract spatial features from both images.
- Combines features using concatenation layers.
- Generates a Stego Image that visually resembles the cover image.
- Optimized using reconstruction loss and similarity metrics.

The encoder minimizes the difference between:

- Cover Image and Stego Image
- Secret Image and Reconstructed Image

This ensures imperceptibility and accurate recovery.

3. Decoder Network for Secret Image Extraction

The Decoder extracts the hidden image from the Stego image.

- Takes Stego Image as input.
- Uses multiple convolutional layers to reconstruct the secret image.
- Learns inverse mapping of encoder transformations.
- Minimizes reconstruction loss (MSE).

The decoder ensures that the extracted image closely matches the original secret image.

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4. Attention Mechanism for Improved Embedding

An Attention Mechanism is integrated into the network to improve feature selection.

- Assigns higher importance to informative regions of the image.
- Helps reduce unnecessary distortion in smooth areas.
- Improves reconstruction accuracy of secret image.
- Enhances robustness against noise and compression attacks.

The attention layer improves the overall security and performance of the steganography model

5. Loss Functions Used

To optimize the model, multiple loss functions are applied:

- **Reconstruction Loss (MSE)** – Measures difference between original and recovered secret image.
- **Cover Loss** – Measures difference between cover and stego image.
- **Total Loss** – Weighted combination of both losses.

This ensures balance between invisibility and extraction accuracy.

To implement this project we have designed following modules:

1) Upload Cover & Secret Image Dataset:

Using this module we will upload cover images and secret images dataset to the application for training and testing.

2) Dataset Preprocessing:

Using this module we will read all images, resize them into equal size, normalize pixel values, convert into array format and shuffle the dataset.

3) Run Proposed CNN Encoder–Decoder Algorithm:

Using this module we will split dataset into 80% training and 20% testing data. Training data will be given to CNN Encoder–Decoder model where encoder hides secret image into cover image and decoder extracts it back. Then trained model will be applied on test data to calculate PSNR, SSIM, MSE and reconstruction accuracy.

4) Run Attention-Based CNN Encoder–Decoder Algorithm:

In this module we will integrate attention mechanism with CNN Encoder–Decoder model to improve embedding quality. After training and testing, performance metrics will be calculated.

Image Steganography with CNN Based Encoder-Decoder Model

5) Run Existing LSB Algorithm:

Using this module we will implement traditional LSB steganography method and calculate PSNR and MSE values for comparison.

6) Comparison Graph:

Using this module we will plot comparison graph between Proposed CNN Encoder–Decoder, Attention-Based Model and Existing LSB method based on performance metrics.

7) Secret Image Hiding & Extraction from Custom Input:

Using this module we will upload any custom cover and secret image, generate stego image and extract hidden image to verify secure embedding.

4.2 SAMPLE CODE

views.py

```
from django.shortcuts import render
from django.core.files.storage import FileSystemStorage
import numpy as np
import tensorflow as tf
from tensorflow.keras.preprocessing import image
import matplotlib.pyplot as plt
import os
import cv2
from tensorflow.keras import layers, Model

# Create your views here.
def index(request):
    return render(request, 'index.html')

def LogAction(request):
    uname=request.POST['username']
    pwd=request.POST['password']
    if uname == 'Admin' and pwd == 'Admin':
        return render(request, 'AdminHome.html')
    else:
        context={'msg':'Login Failed...!!'}
        return render(request, 'AdminApp/index.html', context)
```

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```
def home(request):
    return render(request, 'AdminHome.html')

def encode(request):
    return render(request, 'Encode.html')

# Define the character set and mappings for one-hot encoding
char_set=
'ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz0123456789'
char_to_index = {char: idx for idx, char in enumerate(char_set)}
index_to_char = {idx: char for idx, char in enumerate(char_set)}

# Function to convert text to one-hot encoding
def text_to_one_hot(text, char_to_index, text_length):
    one_hot_encoded = np.zeros((text_length, len(char_to_index))) # Shape: (text_length,
vocab_size)
    for i, char in enumerate(text):
        if i < text_length:
            index = char_to_index.get(char, -1) # Get index of the character
            if index != -1:
                one_hot_encoded[i, index] = 1 # Set the corresponding index to 1
    return one_hot_encoded

# Encoder Network: Embeds the secret text into the image
def build_encoder(input_image_shape, text_length, vocab_size):
    image_input = layers.Input(shape=input_image_shape)
    text_input = layers.Input(shape=(text_length, vocab_size))

    # CNN layers to process the image
    x = layers.Conv2D(32, (3, 3), activation='relu')(image_input)
    x = layers.MaxPooling2D()(x)
    x = layers.Conv2D(64, (3, 3), activation='relu')(x)
    x = layers.MaxPooling2D()(x)
    x = layers.Conv2D(128, (3, 3), activation='relu')(x)
    x = layers.MaxPooling2D()(x)
    x = layers.Flatten()(x)
```


Image Steganography with CNN Based Encoder-Decoder Model

```
# Concatenate the text input with the image features
```

```
x = layers.Concatenate()([x, layers.Flatten()(text_input)])
```

```
# Fully connected layer to produce the final encoded image
```

```
encoded_image = layers.Dense(np.prod(input_image_shape), activation='sigmoid')(x)
```

```
encoded_image = layers.Reshape(input_image_shape)(encoded_image)
```

```
encoder = Model(inputs=[image_input, text_input], outputs=encoded_image)
```

```
return encoder
```

```
# Decoder Network: Extracts the secret text from the image
```

```
defbuild_decoder(input_image_shape, text_length, vocab_size):
```

```
image_input = layers.Input(shape=input_image_shape)
```

```
# CNN layers to process the encoded image and extract text features
```

```
x = layers.Conv2D(128, (3, 3), activation='relu')(image_input)
```

```
x = layers.MaxPooling2D()(x)
```

```
x = layers.Conv2D(64, (3, 3), activation='relu')(x)
```

```
x = layers.MaxPooling2D()(x)
```

```
x = layers.Conv2D(32, (3, 3), activation='relu')(x)
```

```
x = layers.MaxPooling2D()(x)
```

```
x = layers.Flatten()(x)
```

```
decoded_text = layers.Dense(text_length * vocab_size, activation='softmax')(x)
```

```
decoded_text = layers.Reshape((text_length, vocab_size))(decoded_text)
```

```
decoder = Model(inputs=image_input, outputs=decoded_text)
```

```
return decoder
```

```
global filename, uploaded_file_url
```

```
defEncodeAction(request):
```

```
global filename, uploaded_file_url
```

```
decoded_text='null'
```

```
ifrequest.method == 'POST' and request.FILES['file']:
```

```
myfile = request.FILES['file']
```

```
message = request.POST['message']
```

```
iflen(message) == 20:
```

```
CMRTC
```

Image Steganography with CNN Based Encoder-Decoder Model

```
fs = FileSystemStorage()
location = myfile.name
filename = fs.save(myfile.name, myfile)

uploaded_file_url = fs.url(filename)

BASE_DIR = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
imgdisplay = cv2.imread(BASE_DIR + "/" + uploaded_file_url)

# Load an image from your system (ensure the path is correct)
#image_path = myfile # Replace with the actual path to your image
img = image.load_img(BASE_DIR + "/" + uploaded_file_url, target_size=(32, 32))
# Resize to 128x128
img_array = image.img_to_array(img) # Convert image to numpy array
# Normalize the image to [0, 1]
img_array = img_array / 255.0

# Display the image
plt.imshow(img_array)
plt.title('Uploaded Image')
plt.show()
plt.close()
print("Image Shape:", img_array.shape)

# Example secret text and text length
secret_text = message
text_length = 20 # Maximum length of the secret text
encoded_text = text_to_one_hot(secret_text, char_to_index, text_length)
print("Encoded Text Shape:", encoded_text.shape)

# Define input image shape
input_image_shape = (32, 32, 3)

# Build encoder and decoder
encoder = build_encoder(input_image_shape, text_length, len(char_to_index))
decoder = build_decoder(input_image_shape, text_length, len(char_to_index))

# Define the full model (encoder + decoder)
encoded_image = encoder.output
```

CMRTC

Image Steganography with CNN Based Encoder-Decoder Model

```
decoded_text = decoder(encoded_image)
full_model = Model(inputs=[encoder.input[0], encoder.input[1]], outputs=decoded_text)
full_model.compile(optimizer='adam', loss='categorical_crossentropy', metrics=['accuracy'])
full_model.summary()
```

```
    # Prepare the image and text for training
train_images = np.expand_dims(img_array, axis=0) # Add batch dimension
train_texts = np.expand_dims(encoded_text, axis=0) # Add batch dimension
```

```
    # Train the full model
full_model.fit([train_images, train_texts], train_texts, epochs=25, batch_size=1)
```

```
    # Save the model for future use
full_model.save('encoder_decoder_model.h5')
```

```
    # Test the model with a new image and secret text
test_image = np.expand_dims(img_array, axis=0) # Use the same test image
test_text = np.expand_dims(encoded_text, axis=0) # Use the same secret text
```

```
    # Encode the secret text into the image
encoded_test_image = encoder.predict([test_image, test_text])
```

```
    # Remove the batch dimension by squeezing it (assuming only one image in the batch)
img_single = encoded_test_image[0]
```

```
    # Or use img_array.squeeze() if you want to squeeze all dimensions
    # Display the image
plt.imshow(img_single)# img_single will have shape (32, 32, 3)
plt.title('Steganography Image')
plt.axis('off') # Hide axis for clarity
plt.savefig('Stegano_image.png', bbox_inches='tight', pad_inches=0)
plt.show()
plt.close()
```

Image Steganography with CNN Based Encoder-Decoder Model

```
# Decode the text from the encoded image
decoded_text_from_image = decoder.predict(encoded_test_image)

# Convert the decoded output (one-hot encoding) back to text
decoded_text_from_image = np.argmax(decoded_text_from_image, axis=-1)

# Get the indices
decoded_text = "".join([index_to_char[idx] for idx in decoded_text_from_image[0]])
print("Decoded Text:", decoded_text)
context = {'data': decoded_text}
return render(request, 'Decode.html', context)
else:
context = {'data': "Message Length Must Be 20 Characters"}
return render(request, 'Encode.html', context)
```


5. RESULTS & DISCUSSION

Image Steganography with CNN Based Encoder-Decoder Model

5. RESULTS & DISCUSSION

The following screenshots show the results of our project **Image Steganography with CNN-Based Encoder-Decoder Model**, highlighting its key features and functionality. They demonstrate how the system hides secret images inside cover images and successfully retrieves them. These visuals illustrate the performance and effectiveness of the proposed steganography model.

5.1 GUI/Main Interface :

SCREEN SHOTS

>> Index page

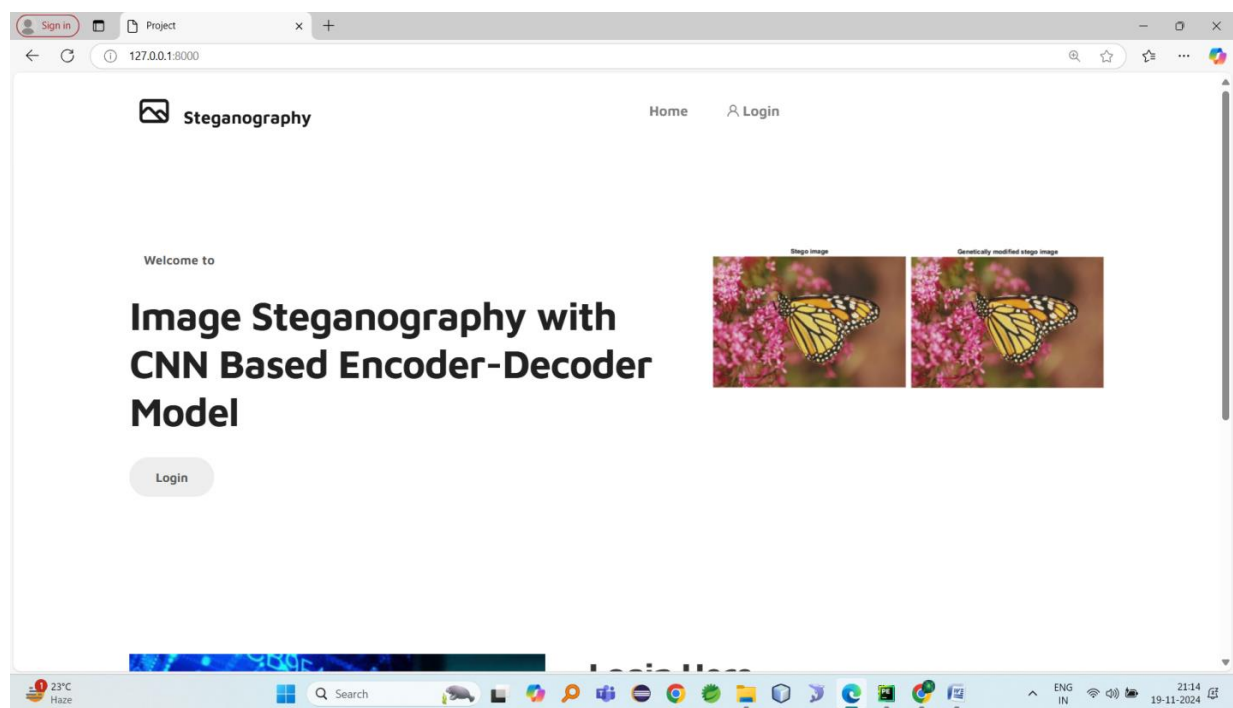


Figure 5.1: User Interface of the Image Steganography System Showing Stego Image Generation and Visualization

Image Steganography with CNN Based Encoder-Decoder Model

>>Admin login

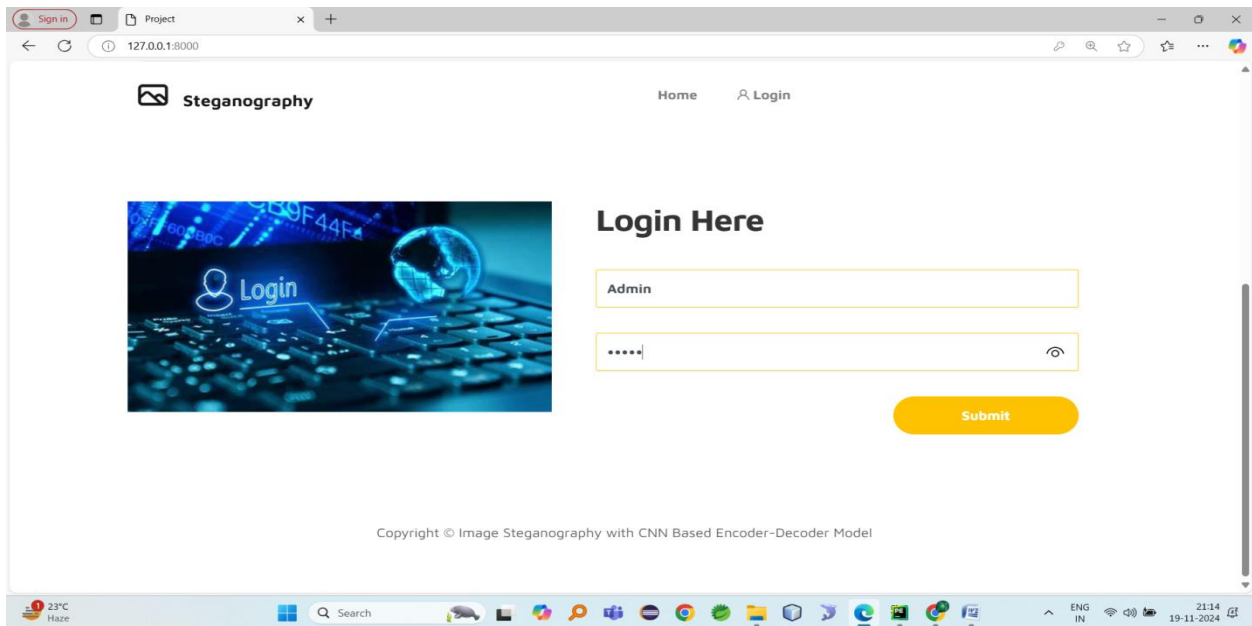


Figure 5.2: User Login Interface of the CNN-Based Steganography System for Administrator Acce

>>Admin home page

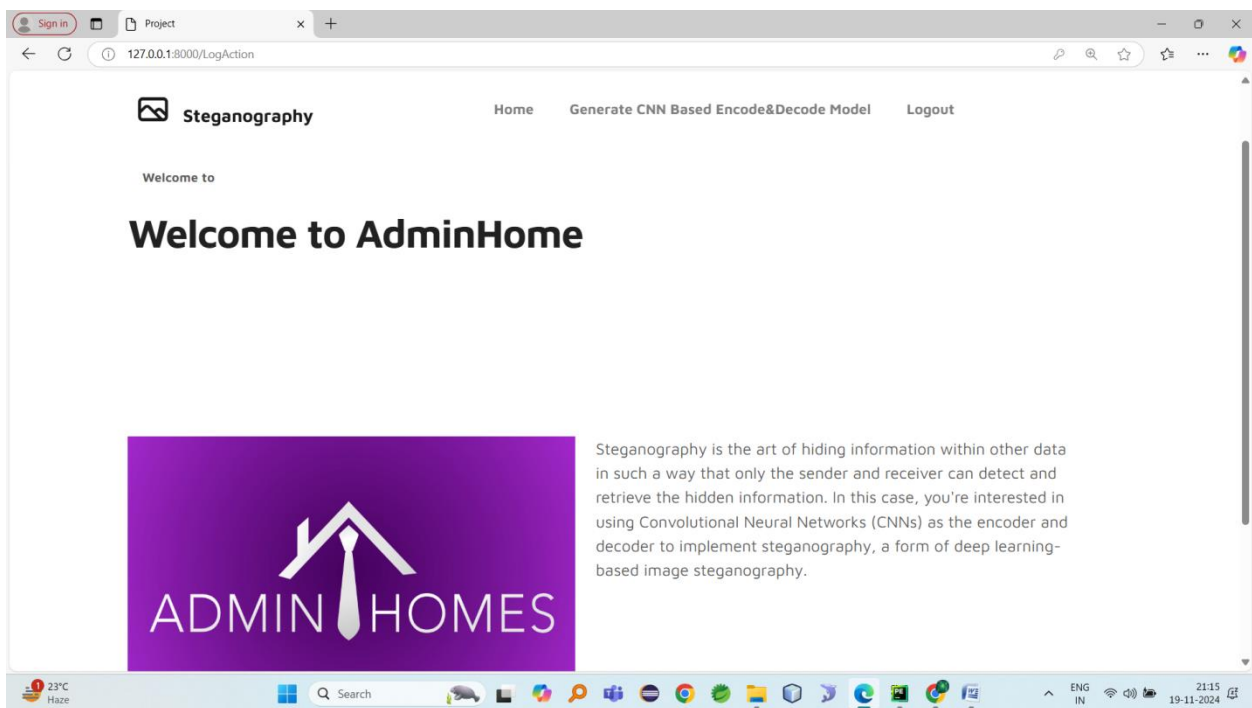


Figure 5.3: Administrator Dashboard Home Page of the CNN-Based Steganography System providing Navigation to Model Generation Modules

Image Steganography with CNN Based Encoder-Decoder Model

>>CNN Based Encode And Decode

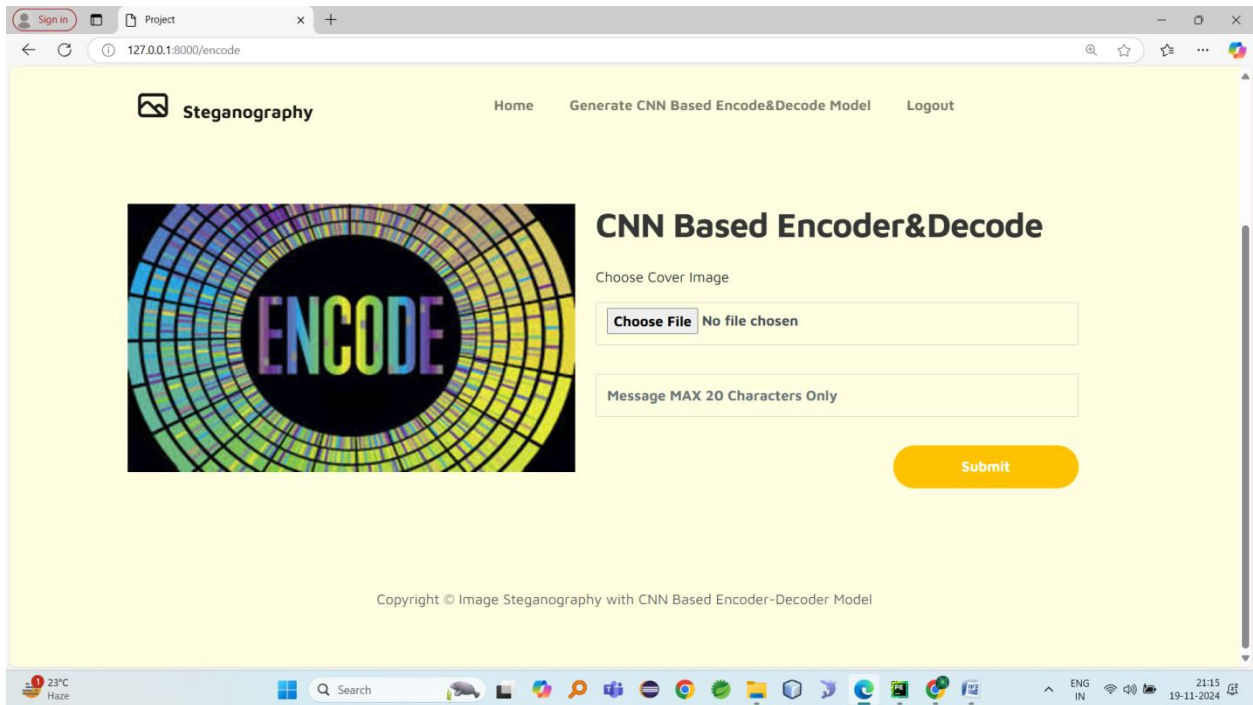


Figure 5.4: Data Embedding Module Interface of the CNN-Based Steganography System for Generating Stego Images

>>Uploading dog image as cover image

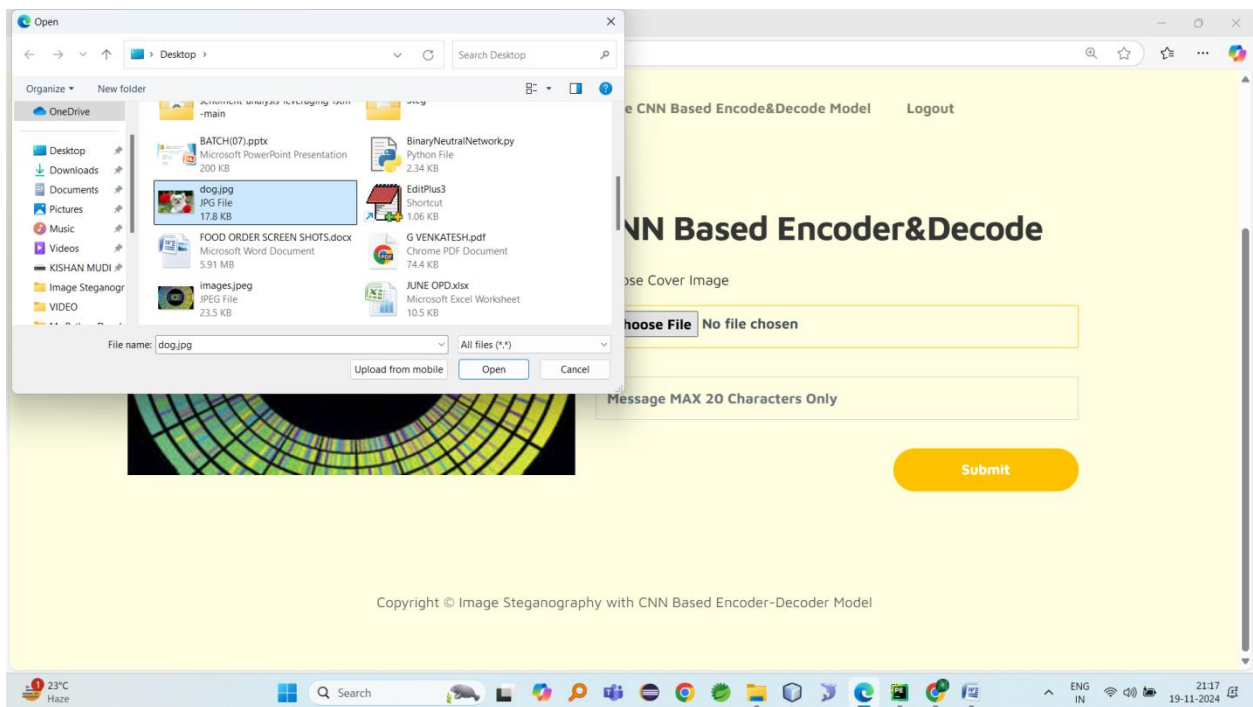


Figure 5.5: Data Selection Interface of the CNN-Based Steganography System for Choosing Cover Images and Inputting Secret Messages

Image Steganography with CNN Based Encoder-Decoder Model

>>And we are trying to add data in to dog image

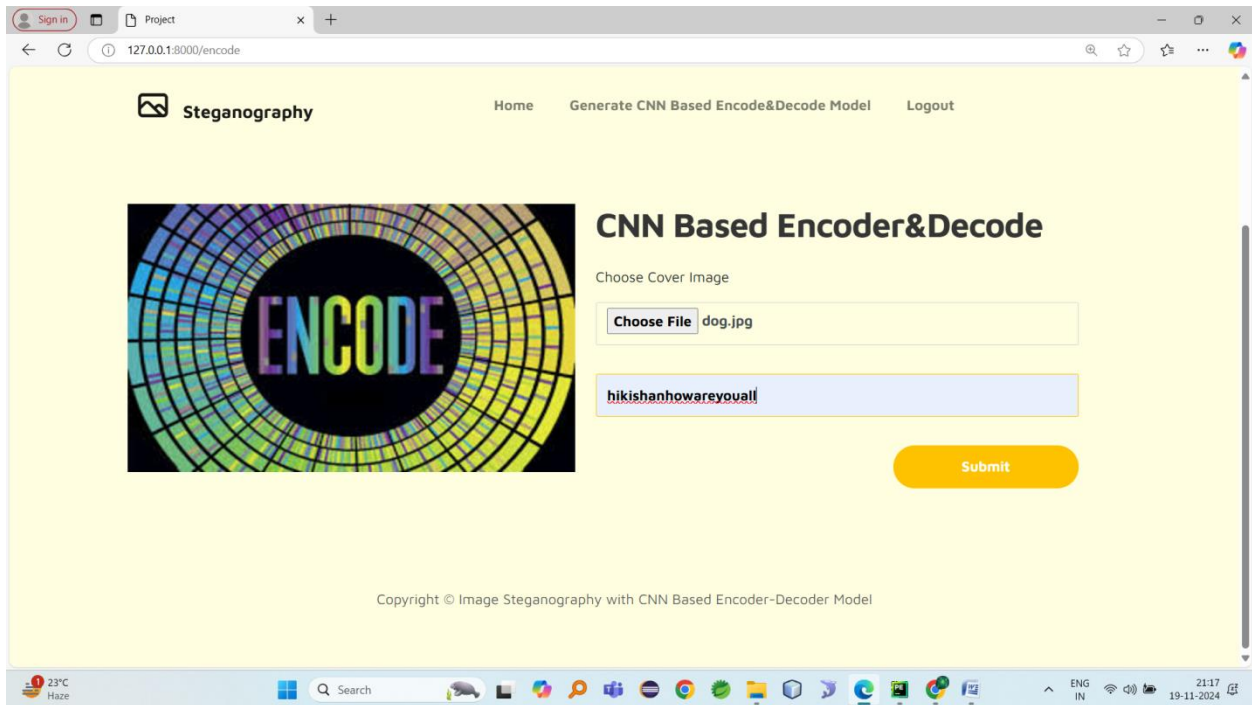


Figure 5.6: Encoding Module Interface of the CNN-Based Steganography System with Selected Cover Image and Secret Message Entry

>>See dog image after pre-process into 32X32 pixels

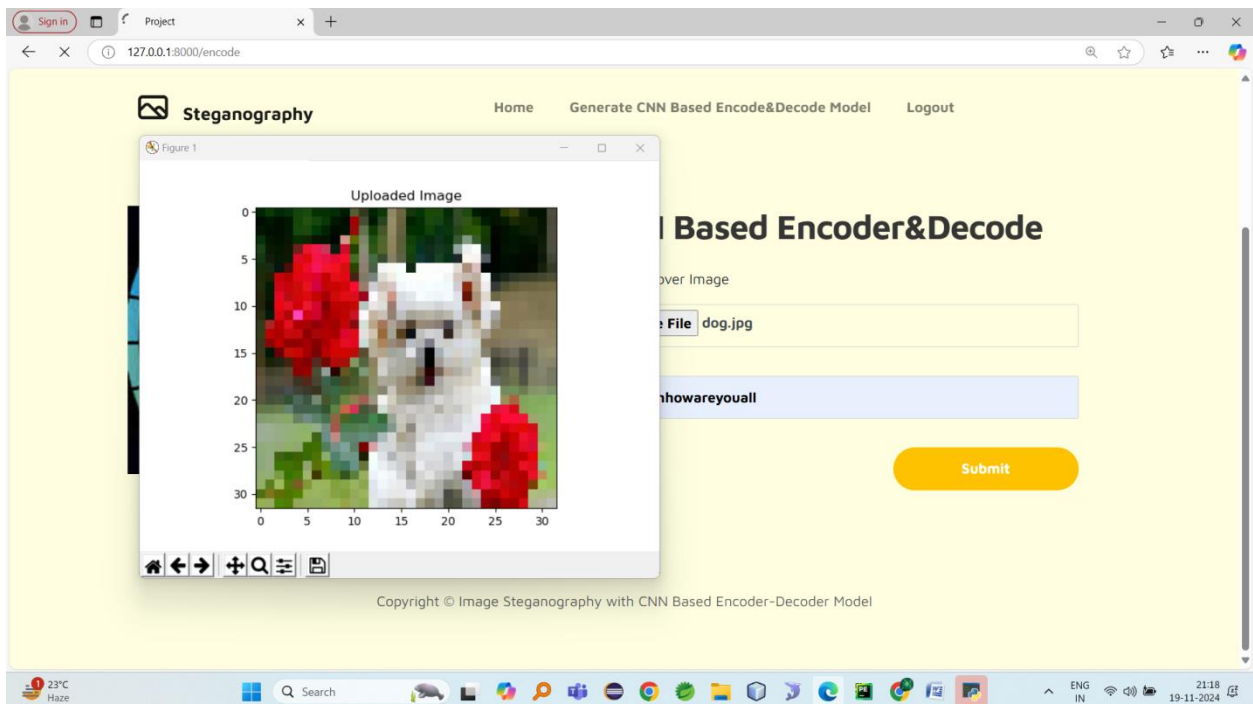


Figure 5.7: Visualization of the Uploaded Cover Image within the CNN-Based Steganography System prior to the Encoding Process

Image Steganography with CNN Based Encoder-Decoder Model

>>CNN based encoded model generated and steganography image generated

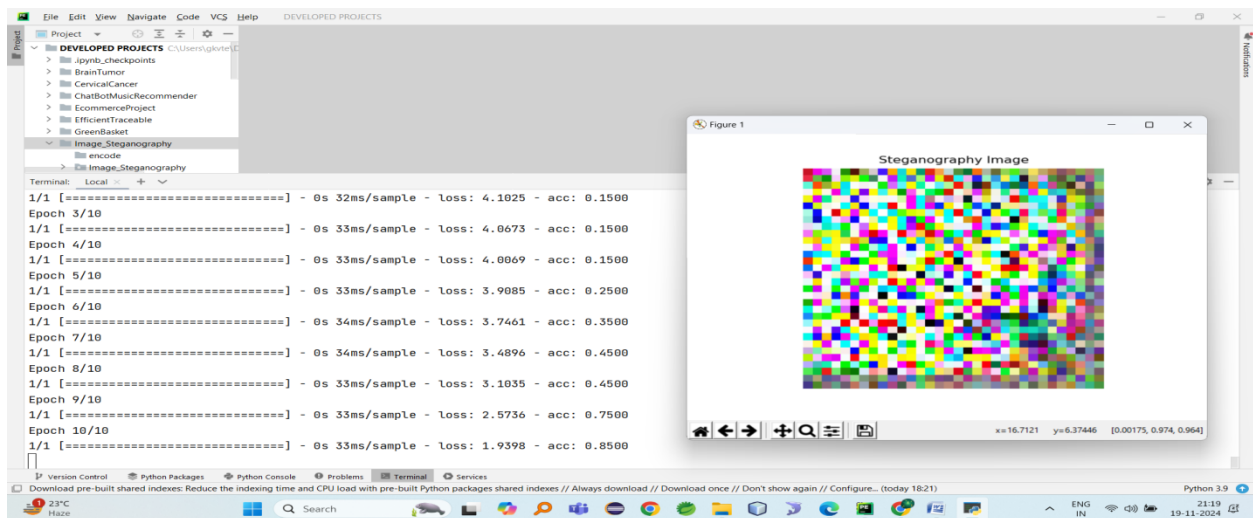


Figure 5.8: Execution of Model Training Epochs showing Loss and Accuracy Metrics alongside the Generated Steganography Image

>>decoded text (90-100% accurate decodes)

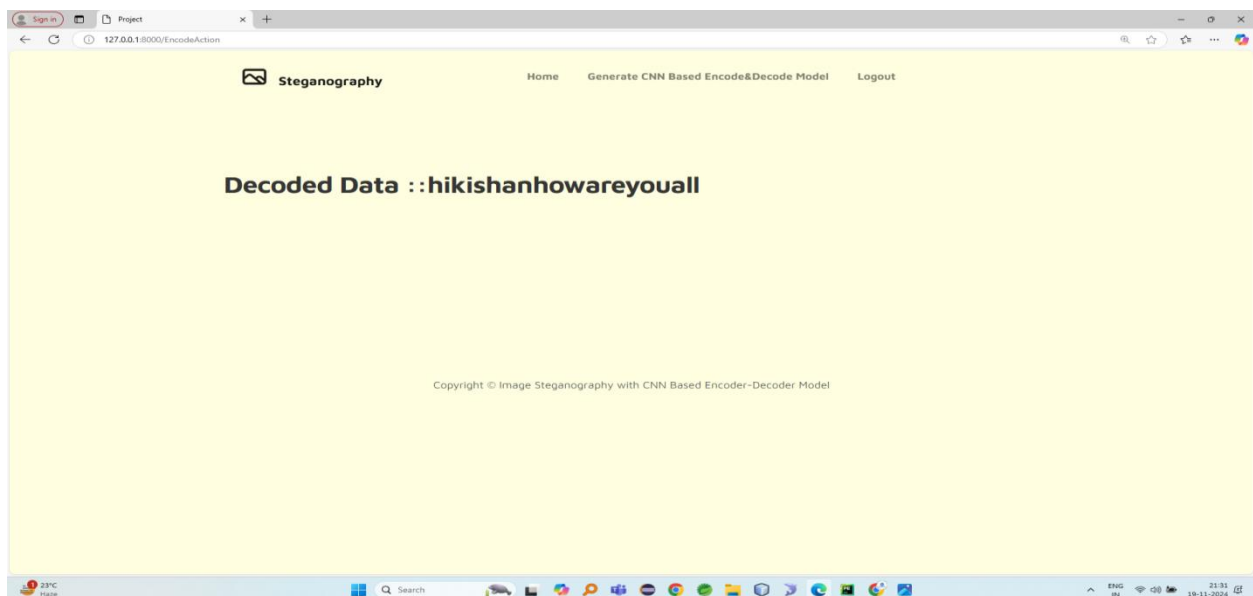


Figure 5.9: Decoding Module Output showing the Successfully Retrieved Secret Message from the Stego Image using the CNN-Based D

6. VALIDATION

6. VALIDATION

The validation phase plays an important role in evaluating the performance and reliability of the proposed **Image Steganography with CNN Based Encoder-Decoder Model**. The main objective of validation is to ensure that the system is capable of securely hiding secret information inside a cover image and accurately retrieving the hidden data without significantly affecting the visual quality of the image. The validation process verifies whether the encoder network successfully embeds the secret information and whether the decoder network is able to reconstruct the hidden data from the generated stego image.

6.1 INTRODUCTION

In this project, the dataset used for training and testing is divided into two parts using an **80:20 ratio**. The training dataset is used to train the CNN-based encoder-decoder model so that it can learn how to effectively hide and retrieve secret data. The testing dataset is then used to evaluate the performance of the trained model on unseen images. This approach helps to measure the generalization ability of the model and ensures that it performs effectively in real-world scenarios.

To evaluate the efficiency of the proposed steganography system, several performance metrics are used. These include **Peak Signal-to-Noise Ratio (PSNR)**, **Structural Similarity Index (SSIM)**, and **Mean Squared Error (MSE)**. PSNR measures the similarity between the cover image and the stego image, where higher values indicate better image quality. SSIM measures the structural similarity between the original and reconstructed images, and MSE calculates the average difference between pixel values of the images.

The experimental validation results demonstrate that the proposed CNN-based model successfully generates stego images that are visually almost identical to the original cover images while accurately reconstructing the hidden secret information. The obtained PSNR and SSIM values indicate that the system maintains high visual quality and ensures reliable extraction of hidden data. These results confirm that the proposed model provides an efficient and secure approach for image steganography.

6.2 TEST CASES

TABLE 6.3.1 UPLOADING DATASET

Test case ID	Test case name	Purpose	Test Case	Output
1	Upload Image Dataset	To verify whether the system correctly loads cover and secret images	User uploads image dataset	Dataset successfully loaded

TABLE 6.3.2 SECRET DATA EMBEDDING

Test case ID	Test case name	Purpose	Input	Output
1	Encode Secret Message	To check whether the encoder hides secret data inside the cover image	Cover image and secret message	Stego image generated successfully
2	Decode Secret Message	To check whether the decoder extracts hidden information from the stego image	Stego image	Secret message retrieved successfully

7. CONCLUSION & FUTURE ASPECTS

7. CONCLUSION & FUTURE ASPECTS

The project “Image Steganography with CNN Based Encoder–Decoder Model” successfully demonstrates a secure and efficient method for hiding secret information within digital images using deep learning techniques. The proposed system uses a Convolutional Neural Network

(CNN)–based encoder–decoder architecture, where the encoder network embeds the secret message into the cover image and the decoder network retrieves the hidden information from the generated stego image.

The implementation of the CNN-based approach provides several advantages compared to traditional steganography methods such as **Least Significant Bit (LSB)** or transform-based techniques. The deep learning model automatically learns complex image features and optimal embedding patterns, which helps in maintaining high visual similarity between the cover image and the stego image. As a result, the hidden data becomes difficult to detect by unauthorized users or steganalysis tools.

7.1 PROJECT CONCLUSION

Overall, the project proves that deep learning-based steganography techniques provide a reliable and secure approach for confidential data transmission. The proposed CNN encoder–decoder model ensures better security, higher embedding capacity, and improved robustness compared to traditional methods, making it suitable for modern cybersecurity and digital communication applications.

7.2 FUTURE ASPECTS

Although the proposed system provides an effective solution for image steganography, there are several opportunities for further improvement and expansion. Future research can focus on enhancing the model performance, improving robustness, and extending the system for more advanced applications.

Future work can also include the development of real-time steganography systems for secure communication over the internet and multimedia platforms. Additionally, lightweight deep learning architectures can be implemented to enable the system to run efficiently on mobile devices, embedded systems, and IoT platforms.

Further research may also focus on improving the robustness of the system against various image processing attacks such as **compression, cropping, noise addition, filtering, and scaling**. By incorporating advanced deep learning techniques and larger datasets, the system can be made more reliable and adaptable for real-world security applications.

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8. REFERENCES

8. REFERENCES

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8.1 GITHUB LINK

<https://github.com/shettyrishit-1707/IMAGE-STEANOGRAPHY->